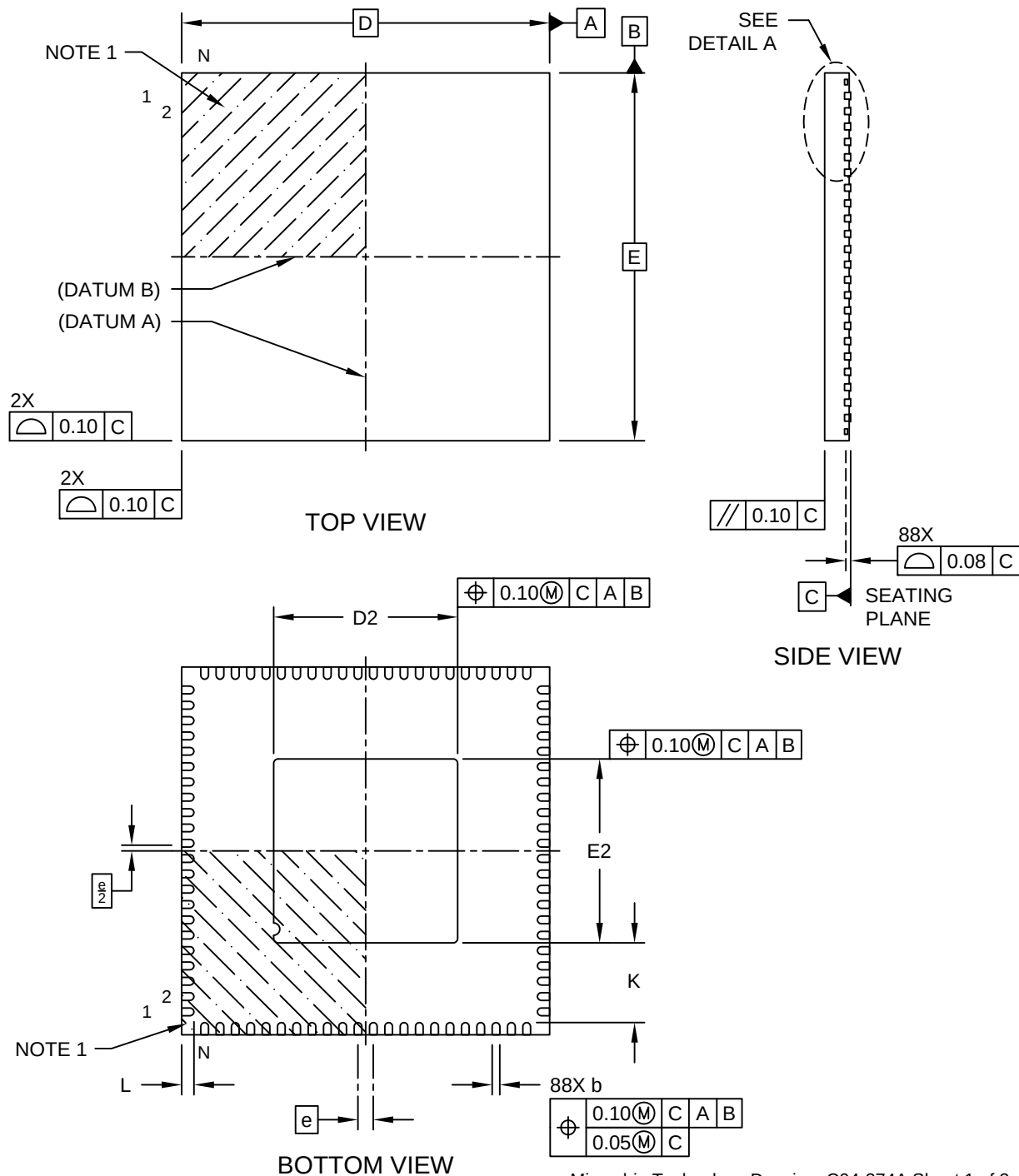


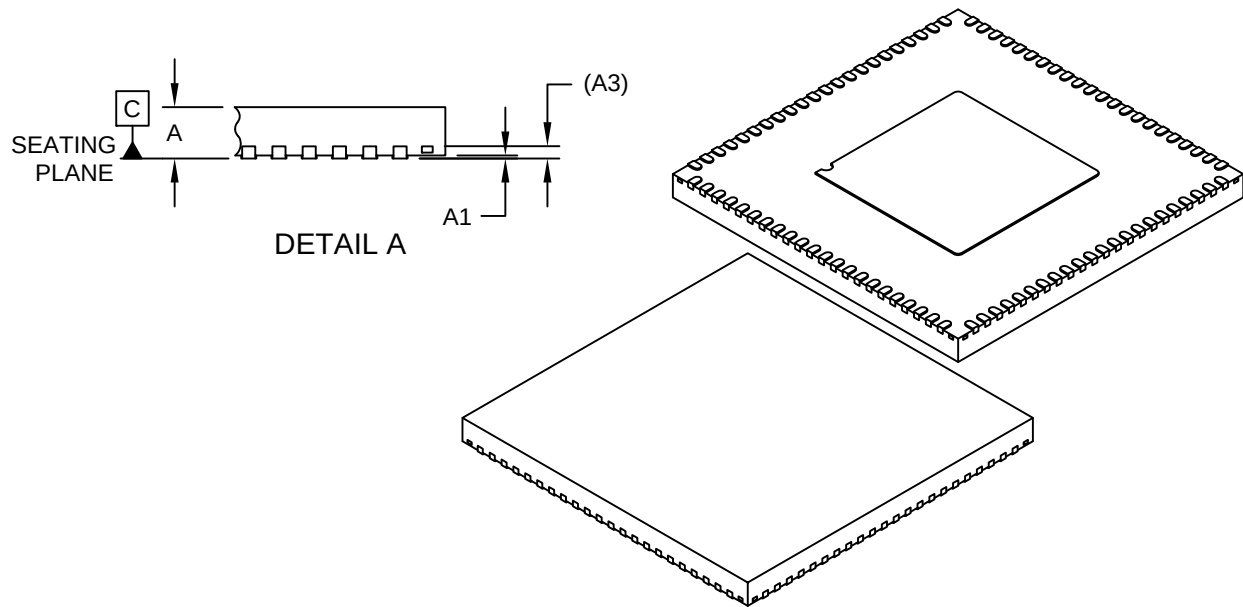
## 88-Lead Very Thin Plastic Quad Flat, No Lead Package (PJ) - 12x12 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 88-Lead Very Thin Plastic Quad Flat, No Lead Package (PJ) - 12x12 mm Body [VQFN]

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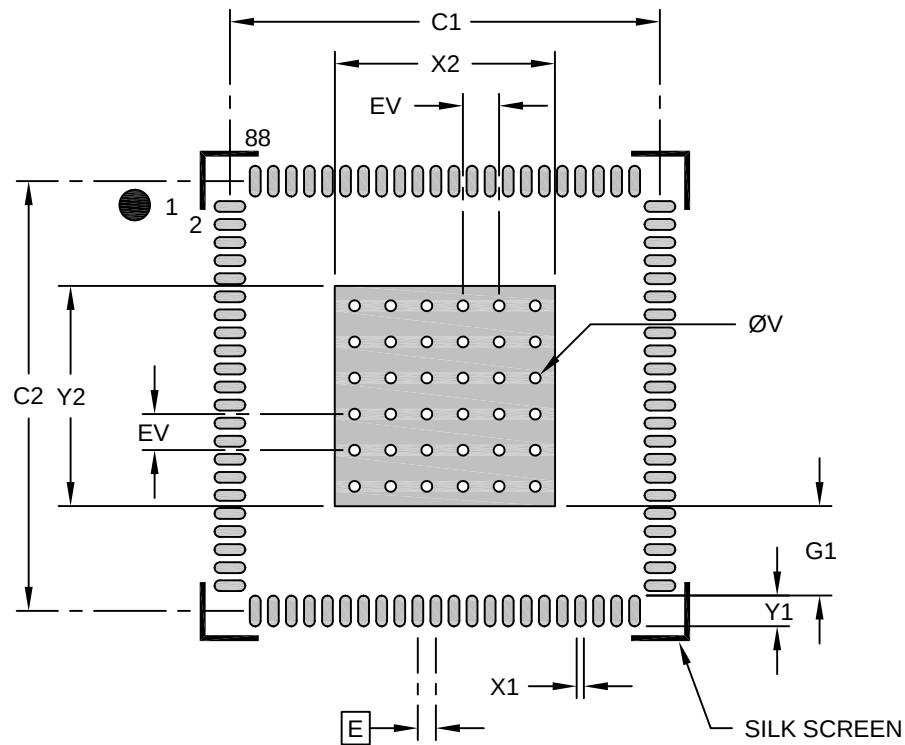
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 88          |      |      |
| Pitch                   | e  | 0.50 BSC    |      |      |
| Overall Height          | A  | 0.80        | 0.85 | 0.90 |
| Standoff                | A1 | 0.00        | 0.02 | 0.05 |
| Terminal Thickness      | A3 | 0.203 REF   |      |      |
| Overall Length          | D  | 12.00 BSC   |      |      |
| Exposed Pad Length      | D2 | 5.90        | 6.00 | 6.10 |
| Overall Width           | E  | 12.00 BSC   |      |      |
| Exposed Pad Width       | E2 | 5.90        | 6.00 | 6.10 |
| Terminal Width          | b  | 0.18        | 0.25 | 0.30 |
| Terminal Length         | L  | 0.30        | 0.40 | 0.50 |
| Terminal-to-Exposed-Pad | K  | 2.50        | -    | -    |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

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### RECOMMENDED LAND PATTERN

| Units                           |    | MILLIMETERS |       |      |
|---------------------------------|----|-------------|-------|------|
| Dimension Limits                |    | MIN         | NOM   | MAX  |
| Contact Pitch                   | E  | 0.50 BSC    |       |      |
| Optional Center Pad Width       | X2 |             |       | 6.10 |
| Optional Center Pad Length      | Y2 |             |       | 6.10 |
| Contact Pad Spacing             | C1 |             | 11.90 |      |
| Contact Pad Spacing             | C2 |             | 11.90 |      |
| Contact Pad Width (X88)         | X1 |             |       | 0.30 |
| Contact Pad Length (X88)        | Y1 |             |       | 0.85 |
| Contact Pad to Center Pad (X88) | G1 | 0.20        |       |      |
| Thermal Via Diameter            | V  |             | 0.30  |      |
| Thermal Via Pitch               | EV |             | 1.00  |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process